A surviving action defect and consistency under inserting cuts

A03, reviewed with B12, 2026-09-07. Two exact tests separate refinement of one experiment from a changing family of fluctuation preparations. Constant-force chord errors vanish on every mesh tending to zero. Within the scalar Gaussian bridge family, exact consistency under retaining old nodes fixes the fluctuation parameter. These statements locate the next selection obligation at the refinement law.

1. Arbitrary partitions of the constant-force trajectory

Fix $m, F, T > 0$, $x(t) = v_0 t$ with $v_0 > 0$, $y(t) = Ft^2/(2m)$, and the Lagrangian $L = m(\dot{x}^2 + \dot{y}^2)/2 + Fy$. Let $\pi = (0 = t_0 < \cdots < t_N = T)$, $\tau_j = t_j - t_{j-1}$, and let q_π linearly interpolate the exact trajectory at these times. Both global endpoints and all sampled positions are fixed in this comparison.

On interval j, with $s = t - t_{j-1}$, the interpolation error is $\eta_j(s) = Fs(\tau_j - s)/(2m)$. The classical first variation vanishes on that interval. Adding the exact local quadratic action differences gives

$$D_\pi := S[q_\pi] - S[q_{\text{cl}}] = \frac{F^2}{24m} \sum_j \tau_j^3 \le \frac{F^2 T}{24m} |\pi|^2 \longrightarrow 0.$$

The unsigned chord–arc lens areas similarly add to $A_\pi = v_0 F \sum_j \tau_j^3/(12m)$, so $D_\pi = FA_\pi/(2v_0)$. The existing constant-force calculation applies after subtracting the local linear tangent; its slope cancels from the lens difference.

For fixed N, convexity yields $\sum_j \tau_j^3 \ge T^3/N^2$, attained by equal steps. Splitting one interval into positive lengths a, b reduces the action defect by

$$\frac{F^2}{24m}[(a+b)^3 - a^3 - b^3] = \frac{F^2}{8m} ab(a+b) > 0.$$

Thus nonuniform cuts cannot defeat convergence when the maximum step tends to zero. This calculation concerns sampled chords; Newton's impulsive polygon is a separate approximation scheme requiring its own error estimate.

2. Exact consistency of Gaussian node laws

Fix the free-line experiment on $[0, T]$, mass $m > 0$ and fixed endpoints. For each partition with at least one internal node, let the centered node vector be Gaussian with covariance

$$\operatorname{Cov}_\pi(\eta_i, \eta_j) = \frac{\kappa_\pi}{m} G(t_i, t_j), \qquad G(s,t) = \min(s,t) - st/T, \quad \kappa_\pi \ge 0.$$

The zero value denotes a point mass at the classical straight path. Exact refinement consistency means that when $\pi' \supset \pi$, deleting the new coordinates from the finer law gives precisely the coarser law, with the same physical times, positions and mass.

Proposition. Within this family, consistency forces $\kappa_{\pi'} = \kappa_\pi$ for every refinement retaining an interior node. Conversely a fixed κ gives consistent Gaussian node laws.

Proof. At any retained interior time s, equality of the one-node marginals implies $\kappa_{\pi'} s(T - s)/(mT) = \kappa_\pi s(T-s)/(mT)$. The factor multiplying κ is positive. Conversely the restriction of a centered Gaussian vector has the corresponding covariance submatrix, which here is exactly the coarse covariance. On the directed family of all partitions, any two nontrivial partitions have a common refinement, so their parameters agree. The endpoint-only partition has no observable variance and places no extra constraint. This completes the proof.

For nested π_N retaining an interior point s, M05's finite-defect scaling $(N-1)\kappa_N \to \ell > 0$ makes that retained variance approach zero. It therefore describes changing preparations, rather than exact marginals of one positive-width bridge experiment. It remains a valid triangular-array limit, with the action defect proved in C014.

3. What survives at fixed preparation?

Let $d = N - 1$. Under the fixed-$\kappa > 0$ bridge, the polygon's free action excess satisfies $2D_\pi/\kappa \sim \chi^2_d$, independent of the actual positive step lengths (C014's finite law). Consequently

$$\mathbb{E}D_\pi = d\kappa/2, \qquad \operatorname{Var} D_\pi = d\kappa^2/2,$$

and $D_\pi \to \infty$ in probability as $d \to \infty$. Indeed, Chebyshev gives $\Pr\{D_\pi < d\kappa/4\} \leq 8/d$. The normalized estimator

$$\widehat{\kappa}_\pi = \frac{2D_\pi}{d}$$

has mean κ and variance $2\kappa^2/d$, hence converges to κ in mean square, even on nonuniform partitions. It estimates the supplied bridge parameter. It is not an additive accumulated action; division by the number of internal nodes is essential. For $\kappa = 0$ every defect vanishes.

Thus exact consistency permits a fixed positive action parameter and a consistent estimator of it, while the unnormalized kinetic action diverges. It also permits the deterministic zero family. Selection of a positive scale still requires a physical premise excluding that family and fixing the scale.

4. One inserted cut and the fluctuation it introduces

On a coarse interval of length $a + b$, fix endpoint positions x, z and insert y after duration a, with $a, b > 0$. Set $\bar{y} = (bx + az)/(a + b)$ and $\zeta = y - \bar{y}$. Completing the square gives

$$\frac{m}{2}\left[\frac{(y-x)^2}{a} + \frac{(z-y)^2}{b} - \frac{(z-x)^2}{a+b}\right] = \frac{m(a+b)}{2ab}\zeta^2.$$

At fixed bridge parameter κ, conditional Gaussian variance is $\operatorname{Var}(\zeta \mid x, z) = \kappa ab/[m(a+b)]$. The expected extra action from inserting one node is therefore $\kappa/2$, independent of the interval lengths. For deterministic interpolation $\zeta = 0$ it is zero. The kinetic splitting identity is nonnegative, whereas section 1's total constant-force chord error decreases under refinement: the latter includes the potential term and moves the inserted node onto the accelerated classical curve.

This is a local, testable formulation of the missing premise: does inserting a physical cut introduce a nonzero conditional fluctuation, and what law sets its variance? Supplying the Gaussian rule answers the first question by assumption. A classical derivation would have to produce that rule or a different consistent fluctuation mechanism from independent physics.

5. Next test and prior-art gate

The B12 audit classifies C027–C029 as elementary consequences. Pitman and Yor, *A guide to Brownian motion and related stochastic processes* (2018), arXiv:1802.09679v1, printed pp. 6, 10, 12–13, supply the Gaussian bridge and restriction framework. The source companion records coverage. The constant-force identity extends the existing calculation. The action law and moments in section 3 are proved in the regulator-limits paper. The combined selection test is the research use of these established ingredients.

Next distinguish coordinate sampling from a physically executed intervention at a cut. A candidate intervention law must state its effect on energy, momentum, conditional variance and coarse observables. Test consistency first, then positivity, scale universality and compatibility with finite speed. The Gaussian conditional law is a reference model for that audit, not yet a model with a hard physical speed ceiling.